

Important Diagrams for Board Exam & NCERT Page Numbers

Class 11 Biology — Chapter 5: Morphology of Flowering Plants

This reference sheet details all official diagrams from Chapter 5 (NCERT Class 11 Biology), mapped with their exact **Figure Numbers**, **NCERT Page Numbers**, **Exam Importance Level**, and **Key Labelling Points** frequently tested in Board exams.

Master List of Important Diagrams

Fig No.	Diagram Name	NCERT Page	Exam Priority	Key Labels / Sub-types to Practice
Figure 5.3	The regions of the root-tip	Page 59	★★★ Very High	Root cap, Region of meristematic activity, Region of elongation, Region of maturation, Root hair
Figure 5.15	Structure of a monocotyledonous seed (Maize grain L.S.)	Page 67	★★★ Very High	Seed coat & fruit-wall, Aleurone layer, Endosperm, Scutellum, Coleoptile, Plumule, Radicle, Coleorhiza
Figure 5.14	Structure of dicotyledonous seed (Gram seed)	Page 66	★★★ Very High	Seed coat, Testa, Tegmen, Hilum, Micropyle, Embryonal axis, Cotyledons, Plumule, Radicle
Figure 5.11	Types of aestivation in corolla	Page 64	★★★ Very High	(a) Valvate, (b) Twisted, (c) Imbricate, (d) Vexillary (Standard, Wings, Keel)
Figure 5.12	Types of placentation	Page 65	★★★ Very High	(a) Marginal, (b) Axile, (c) Parietal, (d) Free central, (e) Basal (with examples)
Figure 5.16 & 5.17(f)	Floral diagram & Floral formula	Page 67, 68	★★★ Very High	Mother axis dot, Calyx, Corolla, Androecium (epipetalous / diadelphous), Gynoecium (locules, placentation), Solanaceae formula
Figure 5.9	Position of floral parts on thalamus	Page 62	★★☆ High	(a) Hypogynous (superior ovary), (b, c) Perigynous (half-inferior ovary), (d) Epigynous (inferior ovary)
Figure 5.13	Parts of a fruit (Mango & Coconut drupe)	Page 66	★★☆ High	Epicarp, Mesocarp (fleshy in mango, fibrous in coconut), Endocarp (stony), Seed
Figure 5.4	Structure of a leaf & Venation	Page 60	★★☆ High	Leaf base, Petiole, Lamina, Stipule, Axillary bud, Reticulate venation, Parallel venation

Fig No.	Diagram Name	NCERT Page	Exam Priority	Key Labels / Sub-types to Practice
Figure 5.5	Compound leaves	Page 60	★☆☆ Moderate	(a) Pinnately compound (Rachis, Neem), (b) Palmately compound (Silk Cotton)
Figure 5.6	Different types of phyllotaxy	Page 61	★☆☆ Moderate	(a) Alternate (China rose), (b) Opposite (Guava/Calotropis), (c) Whorled (Alstonia)
Figure 5.1 & 5.2	Parts of a flowering plant & Root types	Page 58	★☆☆ Moderate	Shoot system, Root system, Tap root, Fibrous root, Adventitious roots

💡 High-Yield Board Exam Drawing Tips:

- **Direct Exercise Questions:** Exercise Q5 explicitly asks to "Draw the labelled diagram of: (i) gram seed (Page 66) (ii) V.S. of maize seed (Page 67)" and Q6 requires drawing the "Floral diagram of Solanaceae (Page 68)".
- **Schematic Diagrams:** Schematic drawings for **Aestivation (Fig 5.11)** and **Placentation (Fig 5.12)** are frequently asked as 3-mark and 5-mark short questions accompanied by definitions and examples.
- **Labelling Accuracy:** For the monocot seed (Maize), ensure proper demarcation of the single cotyledon (**Scutellum**) and protective sheaths (**Coleoptile & Coleorrhiza**).